Linear programming and duality

Optimization · University · Lecture 1



In this lecture

- Formulate LPs.
- See why corners are optimal.
- Interpret duals.

Remember

- Optima at vertices.
- Every LP has a dual.
- Shadow prices value resources.

Draw it, then price it

1. Warm-up (5 min): Beat the staff
2. Draw (20 min): Two variables.
3. Dual (15 min): Write it.
4. Price (15 min): Interpret.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. Which corner won?
2. What does a zero shadow price mean?
3. When does a price stop being valid?

Exit ticket

- Explain one shadow price in plain words.